

Tomer Weiss

Assistant Professor

Department of Informatics
New Jersey Institute of Technology, Newark, New Jersey
☎ +1 (973) 596 6336
✉ tomerwei.github.io
tweiss@njit.edu

I. Education

- 2013–2018 **Ph.D., Computer Science**, *University of California, Los Angeles*.
2013–2015 **M.S., Computer Science**, *University of California, Los Angeles*.
2009–2012 **B.S., Computer Science**, *Tel Aviv University*, Tel Aviv, Israel.

II. Research Interests

- Graphics and Vision** physics-based simulation and modeling, multi-agent dynamics, scene synthesis, gpu algorithms, virtual/mixed reality
Machine Learning statistical and crowd sourced learning, deep reinforcement learning, real-time optimization

III. Experience

A. Academic Appointments

- 2020–Present **Assistant Professor**,
Department of Informatics, Ying Wu College of Computing,
New Jersey Institute of Technology.
Newark, NJ
2019 **Researcher in Residence**,
Institute for Computational and Experimental Research in Mathematics, (ICERM),
Spring Semester Program on Computer Vision,
Brown University.
Providence, RI

B. Non-Academic Employment

- 2018–2019 **Computer Vision Researcher**, *Wayfair*, Boston.
Sum. 2017 **Researcher**, *Autodesk Research*, San Francisco.
Sum. 2016 **Machine Learning Researcher**, *A9.com (Amazon Search)*, Palo Alto.
Sum. 2015 **Software Engineer**, *Bloomberg LP*, New York.
2012–2013 **Software Engineer**, *Parametric Technology Corp.*
2011–2012 **Software Engineer**, *Datonics LLP*.

IV. Awards and Honors

- 2025 **Best Presentation Award**, ACM SIGGRAPH Symposium on Animation, Vancouver, BC, Canada
2025 **Best Student Paper Award**, ACM SIGGRAPH Symposium on Interactive 3D Graphics and Games, Jersey City, NJ
2017 **Best Paper Award**, 10th International ACM SIGGRAPH Conference on Motion in Games, Barcelona, Spain

- 2018 Internationally Selected Invitee, ACM SIGGRAPH ASIA Doctoral Consortium, Tokyo, Japan
- 2018 Finalist, ACM SIGGRAPH Thesis Fast Forward, Vancouver, Canada
- 2016-2017 Graduate Division Fellowship, UCLA
- 2016-2017 Honorarium Award for Service, Computer Science Department, UCLA
- 2013-2014 Doctoral Student Fellowship, Computer Science Department, UCLA

V. Publications

Dashed underlines: students who were/are my advisees

A. Dissertations

- 2018 **A Constraint-Based Approach to Crowd Simulation and Layout Design**
T.Weiss
 Doctoral Thesis, UCLA
- 2016 **Make it Float: Optimizing Design of Physical Objects for Fabrication**
T.Weiss
 Master's Thesis, UCLA

C. Book Chapters

Under Preperation

- 2026 **Fast Simulation of Extremely Massive High-Density Crowds**
T.Weiss
 GPU Zen 5, Black Cat Publishing

Published

- 2024 **Real-Time Simulation of Massive Crowds**
T.Weiss
 GPU Zen 3, Black Cat Publishing, Encinitas, CA.
 ISBN: 979-8344236797
- 2020 **Style Similarity as Feedback for Product Design**
 M.Schwartz, T.Weiss, E.Ataer-Cansizoglu, and J.Choi,
 Morphological Analysis of Cultural DNA, Springer, New York.
 Published: September 24, 2020, pp 27—42, ISBN: 978-981-15-7707-9

D. Refereed Journal Articles

Under Preperation, Review, or Revision

- 2026 Projective Multi-Agent Dynamics for Crowd Simulation
T.Weiss and colleagues
 Computers & Graphics, major revision.
- 2026 Extreme Density Crowd Simulation
T.Weiss and colleagues
 Eurographics 2026
- 2026 ADMM for Crowd Simulation
T.Weiss and colleagues
 ACM SIGGRAPH Interactive 3D Graphics and Games (I3D) '26

Published

- 2026 **Visual clutter: The role of item quantity, spatial organization, and surface complexity in model and human estimates**
 S.Gopu, T.Weiss, and Y. Semizer
 Journal of Vision, Vol. 26, Issue 7, pp 1–13. Published: July 21, 2026 , 2026

- 2025 **Simulating Ant Aggregation Dynamics**
M.Loges and T.Weiss
Proceedings of the ACM on Computer Graphics and Interactive Techniques²
Vol. 8, No. 4, Article 55, pp 1–16. Published: August 8, 2025
Best Presentation Award
- 2025 **Foveated Animations for Efficient Crowd Simulation**
Florin-Vladimir Stancu, T.Weiss, and Rafael Kuffner dos Anjos
Proceedings of the ACM on Computer Graphics and Interactive Techniques¹
Vol. 7, Issue 1, Article 9, pp 1–14. Published: May 22, 2025
Best Student Paper Award
- 2025 **Generalized, Dynamic Multi-agent Torso Crowds**
B.Talukdar and T.Weiss
ACM Proceedings of the ACM on Computer Graphics and Interactive Techniques¹
Vol. 7, Issue 1, Article 4, pp 1–15. Published: May 22, 2025
- 2024 **Learning Crowd Motion Dynamics with Crowds**
B.Talukdar, Y.Zhang, T.Weiss
Proceedings of the ACM on Computer Graphics and Interactive Techniques¹
Vol. 7, Issue 1, pp 1–17. Published: May 13, 2024
- 2023 **Formation-Aware Planning and Navigation with Corridor Shortest Path Maps**
R.Sharma, T.Weiss, and M.Kallmann
Computer Graphics Forum, Vol. 43, Issue 1, February 2024. Published: December 4, 2023
- 2023 **Fast Position-based Multi-Agent Group Dynamics**
T.Weiss
Proceedings of the ACM on Computer Graphics and Interactive Techniques¹
Vol. 6, Issue 1, pp 1–15. Published: May 16, 2023
- 2022 **Reconstructing room scales with a single sound for augmented reality displays**
B.Liang, A.Liang, I.Roman, T.Weiss, B.Duinkharjav, J.Bello, Q.Sun
Journal of Information Display, 24(1), 1–12. Published: Nov. 15, 2022
- 2020 **Image-Driven Furniture Style for Interactive 3D Scene Modeling**
T.Weiss, I.Yildiz, N.Agarwal, E.Cansizoglu, J.Choi
Computer Graphics Forum (Proc. of Pacific Graphics), Vol. 39, Issue 7, pp. 57–68.
Published: Nov. 24, 2020
- 2019 **Fast and Scalable Position-Based Layout Synthesis**
T.Weiss, A.Litteneker, N.Duncan, M.Nakada, C.Jiang, L.Yu, D.Terzopoulos
Short version in IEEE Conference on Computer Vision and Pattern Recognition (CVPR)
workshop on Vision Meets Cognition, June 18–22 2018
IEEE Transactions on Visualization and Computer Graphics (TVCG), Vol. 25, Issue 12, pp.
3231–3243. Published: Dec. 1, 2019
- 2018 **Deep Learning of Biomimetic Sensorimotor Control for Biomechanical Human Anima-
tion**
M.Nakada, T.Zhao, H.Chen, T.Weiss, D.Terzopoulos
ACM Transactions on Graphics (SIGGRAPH), Vol. 36, Issue 4, Article 56, pp 1–15,
Published: Jul. 30, 2018

¹Special issue published as a journal paper of ACM SIGGRAPH Interactive 3D Graphics and Games (I3D)

²Special issue published as a journal paper of the ACM SIGGRAPH Symposium of Computer Animation, Vancouver, Canada, August 8–10, 2025

- 2018 **Position-Based Real-Time Crowd Simulation of Large Crowds**
T.Weiss, A.Litteneker, C.Jiang, D.Terzopoulos
 Computers & Graphics, Vol. 78, pp.12–22. Special Section on *Motion in Games 2017*,
 Published: Oct. 31, 2018
- E. Peer Reviewed Conference Publications**
- 2025 **Projective Multi-Agent Dynamics**
T.Weiss
 Proceedings of the ACM SIGGRAPH Conference on Motion, Interaction, and Games (MIG),
 19: 1–8, Zurich, Switzerland, Dec 3-5
- 2024 **Crowd-sourced Evaluation of Combat Animations**
Y.Zhang and T.Weiss
 IEEE International Conference on Artificial Intelligence and eXtended and Virtual Reality
 (AIxVR), pp. 60–65, Los Angeles, CA, Jan. 17-19
- 2022 **LayoutEnhancer: Generating Good Indoor Layouts from Imperfect Data**
 K.Leimer, P.Guerrero, T.Weiss and P.Musialski
 ACM SIGGRAPH Asia Conference Papers, 27:1–8, Daegu, South Korea, Dec. 6-9
- 2020 **Plane-Based Local Behaviors for Multi-Agent 3D Simulations with Position-Based Dynamics**
 R.Sharma, T.Weiss, and M.Kallmann
 IEEE AIVR 2020, MARCH Workshop: Modeling and Animating Realistic Crowds and Humans,
 pp. 214–217, Virtual, Dec. 14-18
- 2020 **Interactive and Scalable Layout Synthesis with Design Templates**
H.Farooki, E.Ataer-Cansizoglu, J.Choi, T.Weiss
 IEEE AIVR 2020, Industry-track paper, pp. 235–238, Virtual, Dec. 14-18
- 2020 **Interactive Design of Gallery Walls via Mixed Reality**
 H.Huang, Y.Zhang, T.Weiss, R.Perry, L.Yu
 IEEE AIVR 2020, Session on AI for new interaction, experiences, and applications in VR and
 AR, pp. 17–26, Virtual, Dec. 14-18
- 2019 **Room Style Estimation for Style-Aware Recommendation**
 E.Cansizoglu, H.Liu, T.Weiss, A.Mitra, D.Dholakia, J.Choi, D.Wulin
 IEEE International Conference on Artificial Intelligence & Virtual Reality (AIVR), pp. 267–2673,
 San Diego, CA, Dec. 9-11
- 2017 **Position-Based Multi-Agent Dynamics for Real-Time Crowd Simulation**
T.Weiss, A.Litteneker, C.Jiang, D.Terzopoulos
 Proc. of the Tenth International ACM SIGGRAPH Conference on Motion in Games, 10:1–8,
 Barcelona, Spain, Nov 8-10
 (Media: Gamasutra) Invited to contribute an extended blog version
Best Paper Award
- 2017 **Automated Layout Synthesis and Visualization From Images of Interior or Exterior Spaces**
T.Weiss, M.Nakada, D.Terzopoulos
 IEEE Conference on Computer Vision and Pattern Recognition (CVPR) workshop on Vision
 Meets Cognition, pp. 41-47, Honolulu, Hawaii, July 21-26

F. Peer Reviewed Abstracts and Posters

Published

- 2026 **Bayesian Deep Reinforcement Learning for Multi-Agent Crowds**
B.Talukdar and T.Weiss
North East AI Agents Day, NYC, NY, May 8
- 2025 **Simulating the Mechanics of Ant Swarm Aggregations**
M.Loges and T.Weiss
Proceedings of the Special Interest Group on Computer Graphics and Interactive Techniques Conference Posters, 11:1–2, Vancouver, BC, Canada, August 10–14
Invited to participate in student research competition and technical papers program [Rare]
- 2025 **How to Break Crowd Simulation Algorithms**
A.Chen and T.Weiss
Proceedings of the Special Interest Group on Computer Graphics and Interactive Techniques Conference Talks, 61:1–2, Vancouver, BC, Canada, August 10–14
- 2023 **Position-based Torso Crowds**
B. Talukdar, Y.Zhang, T.Weiss
ACM SIGGRAPH Symposium on Interactive 3D Graphics and Games (I3D) Posters, Philadelphia, PA, May 8-10
- 2023 **Learning to Simulate Crowds with Crowds**
B. Talukdar, Y.Zhang, T.Weiss
ACM SIGGRAPH Posters, 6:1–2, Los Angeles, CA, Aug. 6-10
- 2022 **Visual clutter: The role of background texture, set size, and item organization**
Y. Semizer, T.Weiss
Journal of Vision, Issue 23, Vol. 9 (2023): pp. 4940, Vision Sciences Society Annual Meeting Abstract, St. Petersburg, FL, May 19-24, 2023
- 2022 **Learning Anticipatory Multi-Agent Navigation with Position-Based Deep Reinforcement Learning**
B. Talukdar, T.Weiss
Society for Industrial and Applied Mathematics (SIAM) Conference on Mathematics of Data Science, abstract, San Diego, CA, Sep. 26-30
- 2022 **Comparing Visual Clutter Metrics: What Feature is Important?**
Y.Semizer, T.Weiss
American Psychological Association, Poster, Aug. 4-6
- 2022 **Image-Driven 3D Object Style Estimation**
T.Weiss, I.Yildiz, N.Agarwal, and E.Ataer-Cansizoglu
Society for Industrial and Applied Mathematics (SIAM) Conference on Imaging Science, abstract, Virtual, Mar. 21-25
- 2020 **3D Behaviors for Multi-Agent Simulations with Position-Based Dynamics**
R.Sharma, T.Weiss, M.Kallmann
Short version in ACM SIGGRAPH Symposium on Interactive 3D Graphics and Games, Virtual, Sep. 15-17
- 2019 **Implementing Position-Based Real-Time Simulation of Large Crowds**
T.Weiss
MARCH Workshop on Modeling and Animating Realistic Crowds and Humans, IEEE International Conference on Artificial Intelligence & Virtual Reality (AIVR), pp. 306–3061, San Diego, CA, Dec. 9-11

- 2017 **Position-Based Multi-Agent Dynamics for Real-Time Crowd Simulation**
T.Weiss, A.Litteneker, C.Jiang, D.Terzopoulos
Short version in Proc. of the ACM SIGGRAPH/Eurographics Symposium on Computer Animation, Los Angeles, CA, July 28-30, 27:1–2

G. Published Software

Crowd Simulation Method, Open Source, <https://github.com/tomerwei/pbd-crowd-sim>

H. Unrefereed Publications

- 2022 **ATEK: Augmenting Transformers with Expert Knowledge for Indoor Layout Synthesis**
K.Leimer, P.Guerrero, T.Weiss and P.Musialski
ArXiv

VI. Professional Presentations

Invited Talks

- Nov. 2025 Creating 3D scenes and Populating them with Virtual AI Agents at Scale
Invited Speaker, Graduate Computer Vision Seminar, Cornell Tech, NYC, NY
- Nov. 2025 Simulating the Dynamics of Virtual AI Agents at Scale
Invited Speaker, Computer Graphics Seminar, George Mason University, Arlington, VA
- Nov. 2025 Learning to Simulate AI Agent Dynamics at Scale
Invited Speaker, Department Seminar, Stony Brook University, Stony Brook, NY
- Oct. 2025 Learning to Simulate AI Agent Dynamics at Scale
Invited Speaker, Computer Graphics Seminar, Columbia University, New York, NY
- Feb. 2025 Learning to Simulate Virtual AI Agents at Scale
Invited Speaker, Games Division, University of Utah, Salt Lake City, UT
- Nov. 2024 Learning to Simulate AI Agent Dynamics at Scale
Invited Speaker, Computer Science Department, University of Texas San Antonio, TX
- Dec. 2023 Next-Generation Learning for 3D Scene Synthesis at Scale
Invited Speaker, Computer Science Department, Stevens Institute of Technology, NJ
- Apr. 2023 Learning Deep Representations for Collective Behaviors
Invited Speaker, Institute for Future Technologies, New Jersey Institute of Technology, NJ
- Oct. 2022 Interactive Multi-Agent Dynamics Simulation for Physical and Virtual Worlds
Invited Speaker, Computer Science and Engineering colloquium, University of Connecticut, Storrs, CT
- Sep. 2022 Interactive Multi-Agent Dynamics Simulation for Physical and Virtual Worlds
Invited Speaker, Department of Electrical and Computer Engineering, Stevens Institute of Technology, NJ
- May 2022 Virtual Content Synthesis: From Spaces to Multi-Agent Dynamics
Invited Speaker, Toyota Technological Institute, Chicago
- May 2022 Creating virtual spaces with computational interior design and deep learning
Invited Speaker, Computer Graphics Seminar in the Courant Institute of Mathematical Sciences, New York University, New York
- Jan. 2022 Fast Virtual Content Synthesis: From Spaces to Pedestrian Motion Dynamics
Invited Speaker, Cambridge Mobile Telematics, Boston
- Nov. 2021 Real-Time Virtual Content: From Spaces to Crowd Dynamics
Presentation, MARCH Workshop on Modeling and Animating Realistic Crowds and Humans, IEEE International Conference on Artificial Intelligence & Virtual Reality (AIVR)

- Sep. 2021 Fast Virtual Content Synthesis: From Spaces to Crowds
Invited Speaker, Computer Science and Engineering
UC Riverside
- Jul. 2021 Creating virtual spaces with computational interior design and deep learning
Invited Speaker, Artificial Intelligence Division
USC Information Sciences Institute
- Feb. 2021 Bridging the Gap between Physical and Virtual Reality
Invited Speaker, Computer Science Department
UC Davis
- Oct. 2020 Bridging the Gap between Physical and Virtual Reality
Invited Lecture, SIG Research Center for Computer Graphics
University of Pennsylvania
- Apr. 2020 Bridging the Gap between Physical and Virtual Reality
PIXL seminar talk, Computer Science Department
Princeton University
- Nov. 2019 Bridging the Gap between Physical and Virtual Reality
School of Computer Science Weekly Seminar
Interdisciplinary Center Herzliya (IDC)
- Nov. 2019 Bridging the Gap between Physical and Virtual Reality
Faculty of Engineering Seminar
Bar Ilan University
- Nov. 2019 Wayfair: Finding your perfect home with data and computational interior design
Data Science Initiative
Harvard University
- Oct. 2019 Bridging the Gap between Physical and Virtual Reality
Computer Graphics Group Seminar
UMass Amherst
- May 2019 Bridging the Gap between Physical and Virtual Reality
Computer Graphics Group Seminar
CSAIL, MIT
- Apr. 2019 Bridging the Gap between Physical and Virtual Reality
Guest Lecture
Dept. of Computer and Information Science, Fordham University
- Mar. 2019 Bridging the Gap between Physical and Virtual Reality
Colloquia
Dept. of Computer Science, University of Massachusetts Boston
- Mar. 2019 Bridging the Gap between Physical and Virtual Reality
CS Seminar
Dept. of Computer Science, Illinois Institute of Technology
- Mar. 2019 Bridging the Gap between Physical and Virtual Reality
Research Seminar
Dept. of Computer Science, University of North Carolina at Charlotte
- Mar. 2019 Bridging the Gap between Virtual and Physical Worlds
Research Seminar
Digital Worlds Institute, University of Florida

- Mar. 2019 Position-Based Multi-Agent Dynamics for Real-Time Crowd Simulation
Guest Lecture
Digital Worlds Institute, University of Florida
- Mar. 2019 Powering the Next Generation of Human Learning with Immersive Virtual Environments
Research Seminar
Learning Sciences Cluster, University of Central Florida
- Mar. 2019 Bridging the Gap between Physical and Virtual Reality
Research Seminar
Dept. of Computer Science, University of Central Florida
- Feb. 2019 Bridging the Gap between Physical and Virtual Reality
Colloquium
Dept. of Computer Science, Old Dominion University
- Feb. 2019 Bridging the Gap between Physical and Virtual Reality
Research Presentation
Dept. of Informatics, New Jersey Institute of Technology
- Jan. 2019 Introduction to Functional Programming
Guest Lecture
Dept. of Computer Science, Loyola Marymount University
- Nov. 2018 A fast and scalable approach for simulating multi-agent crowds and for creating virtual layouts
Computer Science Colloquium
Dept. of Computer Science, Dartmouth College
- Oct. 2018 A fast and scalable approach for simulating multi-agent crowds and for creating virtual layouts
Computer Science Colloquium
Dept. of Computer Science, University of Massachusetts Lowell
- Oct. 2018 A fast and scalable approach for simulating multi-agent crowds and for creating virtual layouts
Computer Graphics Colloquium
Dept. of Computer Science, University of Pennsylvania
- Jun. 2018 Simulating virtual crowds with 100,000 agents in real-time on your laptop
Boston Computational Creativity Interest Group
Boston, MA
- Mar. 2018 Simulating virtual crowds with 100,000 agents in real-time on your laptop
Visual Computing and Interaction Seminar
Computer Science, Electrical Engineering, Media Arts and Technology (Jointly)
University of California Santa Barbara
- Mar. 2018 Simulating virtual crowds with 100,000 agents in real-time
Dept. Seminar Series
Dept. of Computer Science, University of Massachusetts Boston
- Mar. 2018 Simulating virtual crowds with 100,000 agents in real-time on your laptop
EECS Technical Seminar Series
Electrical Engineering and Computer Science, University of California Merced
- Feb. 2018 Position-based methods for real-time crowd simulation and modeling applications
Computer Graphics and Visualization group
School of Information and Computer Sciences, University of California Irvine
- Feb. 2018 Position-based multi-agent dynamics for real-time crowd simulation and modeling applications
Center for Visual Computing
Dept. of Computer Science and Engineering, University of California San Diego

Dec. 2017 Position-based multi-agent dynamics for real-time crowd simulation
Graphics and Vision Seminar, hosted by Prof. Daniel Cohen-Or
School of Computer Science, Tel Aviv University

Contributed Conference Presentations

- Dec. 2025 Projective Multi-Agent Dynamics
ACM SIGGRAPH Motion, Interaction and Games, Zurich, Switzerland, December 3-5.
- August 2025 How to Break Crowd Simulation Algorithms
ACM SIGGRAPH Talks, Vancouver, BC, Canada, August 10-14.
- May 2023 Fast Position-based Multi-Agent Group Dynamics
ACM SIGGRAPH Symposium on Interactive 3D Graphics and Games, WA, USA, May 3-5.
- Feb. 2022 Computer Graphics & Vision at the New Jersey Institute of Technology ([Link](#))
Workshop on Sustainable Research Pathways for High-Performance Computing, Sustainable Horizons Institute, Rancho Mirage, CA
- Dec. 2020 Position-Based Real-Time Simulation of Large Crowds ([Media](#))
Workshop on Social Dynamics beyond Vehicle Autonomy, Institute for Pure & Applied Mathematics, UCLA
- Sep. 2020 Fast and Scalable Position-Based Layout Synthesis
ACM SIGGRAPH Symposium on Interactive 3D Graphics and Games, Virtual, Sep. 14-18
- Dec. 2019 Implementing Position-Based Real-Time Simulation of Large Crowds
MARCH Workshop on Modeling and Animating Realistic Crowds and Humans,
IEEE International Conference on Artificial Intelligence & Virtual Reality (AIVR), Dec. 9-11
- Dec. 2018 A Constraint-Based Approach to Crowd Simulation and Layout Design
ACM SIGGRAPH Asia Doctoral Consortium
Tokyo, Japan, Dec 4
- Oct. 2018 Gallery Wall Synthesis and Visualization
International Conference on Predictive Applications and APIs, Boston, MA, Oct. 17
- Aug. 2018 A Constraint-Based Approach to Crowd Simulation and Layout Design
ACM SIGGRAPH Thesis Fast Forward, Vancouver, BC, Canada, Aug 14
- Nov. 2017 Position-Based Multi-Agent Dynamics for Real-Time Crowd Simulation
ACM SIGGRAPH Conference on Motion in Games, Nov 8-10
- Jul. 2017 Position-Based Multi-Agent Dynamics for Real-Time Crowd Simulation
Proc. of the ACM SIGGRAPH/Eurographics Symposium on Computer Animation, July 28-30

Other Presentations

- Mar. 2019 Computing Community Consortium (CCC)
Content Generation for Workforce Training workshop, March 14-15

VII. Teaching Experience

A. Classes Taught

- Fall, Spr. **Computer Graphics for Visual Effects**, Informatics, NJIT.
- 2020-26 Covers current techniques and foundational algorithms in graphics, including: 2D/3D graphics, color, images and image processing, rendering technologies, ray tracing, physics-based simulation, animation, and artificial intelligence methods for virtual worlds. Class includes several assignments and a final project.

Spr. 2022 **AI for Artificial Worlds**, Informatics, NJIT.

Artificial worlds enables creators such as artists, programmers and scientists to realize their creative visions. This class is a seminar-oriented research course covering topics at the intersection of graphics, animation, and AI. This course will explore the recent research in AI, multi-agent simulation, artificial physics, content creation, including data-driven and deep learning methods.

Spr. 2022, **Information Design Techniques**, Informatics, NJIT.

Spr./Fall 2025/6 Introduction to interactive programming and design of graphics applications. Includes: user experience diagramming, feature-centered design, event-driven programming, user interfaces, graphics, animations, and interface integration. Projects focus on cross-platform delivery of web applications using WebGL.

Fall, Spr., **Programming Languages**, CS, UCLA.

Win. 2014-18 Basic concepts in design and use of programming languages, including abstraction, modularity, control mechanisms, types, declarations, syntax, and semantics. Study of several different language paradigms: OCaml, Python, Java, Scheme and Prolog.

Spr. 2017 **Software Engineering**, CS, UCLA.

Structured programming, program specification, program proving, modularity, abstract data types, composite design, software tools, software control systems, program testing, team programming.

Win. 2016 **Software Construction Lab**, CS, UCLA.

Fundamentals of open-source and other software tools and environments, used in upper-division computer science courses and in the software industry.

Fall 2015 **Computer Science Freshman Seminar**, CS, UCLA.

Introduction to principal topics and key ideas in computer science and engineering. Seminar includes assignments, independent study, and writing reports.

B. Curriculum Development

New Courses Developed

Spr. 2022 **AI for Artificial Worlds**, Informatics, NJIT.

New Course Manuals Developed

Spr. 2020 **Computer Graphics for Visual Effects**, Informatics, NJIT.

C. Research and Design Supervision

Graduated Doctoral Students

2020-2024 Yunhao Zhang, NJIT PhD in Information Systems, Crowd-Sourced Learning for Computer Graphics Applications

2021-2026 Bilas Talukdar, NJIT PhD in Information Systems, Perceptual and Geometric advances in Crowd Simulation

Thesis Committees

2026 Kaiheng Zhang, NJIT Urban Systems doctoral student, Multi-scale generative framework for performance-guided architectural

2024 Haotian Yin, NJIT Computer Science doctoral student, Advancing 3D Surface Reconstruction and Parametrization from Implicit

2024 Shen Fan, NJIT Computer Science doctoral student, Optimization of Quality and Efficiency for Reconstruction of Explicit 3D Geometry from Implicit Neural Representations of Geometric Shapes

2023 Bilas Talukdar, NJIT Informatics doctoral student, Deep Reinforcement Learning for Crowd Simulation with Crowds

2023 Li Yi, NJIT Informatics doctoral student, Learning Representations For Effective and Explainable Software Bug Detection and Fixing

2022 Yunhao Zhang, NJIT Informatics doctoral student, Sword fighting motion synthesis

2021 Raina Samuel, NJIT Informatics doctoral student, Android Security: Analysis and Applications

Doctoral Qualifying Exam Committees

- 2025 Huu Phuoc Dang, NJIT Informatics doctoral student, Discrete Latent Representations for Efficient Surrogate-Assisted Robot Morphology Optimization
- 2025 Kaiheng Zhang, NJIT Urban Systems doctoral student, Challenges of Generative Models in Architectural Design for Urban Systems
- 2023 Li Yi, NJIT Informatics doctoral student, Learning Representations For Effective and Explainable Software Bug Detection and Fixing
- 2022 Dahlia Musa, NJIT Informatics doctoral student, Virtual Healthcare Wound Care
- 2022 Bilas Talukdar, NJIT Informatics doctoral student, Learning Anticipatory Multi-Agent Navigation with Position-Based Deep Reinforcement Learning
- 2022 Wenbo Wang, NJIT Informatics doctoral student, QA4GIS: A Novel Approach Learning to Answer GIS Developer Questions with API Documentation
- 2021 Yunhao Zhang, NJIT Informatics doctoral student, Sword Fighting Motion Synthesis
- 2021 Raina Samuel, NJIT Informatics doctoral student, Android Security: Analysis and Applications

Undergraduate Students

- 2026 Anderson Rodriguez, NJIT Computer Science student, Simulating Ant Aggregation Self-Assembly
- 2024-25 Matthew Loges, NJIT Computer Science student, Simulating Ant Aggregations
- 2024 Jeremy Convocar, NJIT Informatics student, Dense Flows in Crowd Simulation
- 2022 Audrey Bhea Obispo, NJIT Informatics student, Visualizing Collective Ant Dynamics
- 2022 Barr Beneli, NJIT Informatics student, Simulating Collective Ant Dynamics
- 2022 Elijah Brick, NJIT Computer Science student, Zebrafish Motion Dynamics Simulation
- 2021-2022 Tejasi Thool, NJIT Digital Design student, Spatial design for Promoting Mental Health and Well-being
- 2021 Anuraag Godavari, NJIT Computer Science student, Parallel and GPU Programming Methods for 3D Crowd Simulation
- 2021 Gianncarlo Carrasco Perez, NJIT Computer Science student, floorplan sketch completion with machine learning
- 2020-2021 Muhammed Waggeh, NJIT Informatics student, ant swarm simulation
- 2020 Hameedullah Farooki, NJIT Computer Science student, computational methods for architectural layout synthesis
- 2020 Donovan Westwater, NJIT Informatics student, scalable crowd simulation methods in games

High School Students

- 2025-26 Derek You, Creating social crowd simulation scenarios for learning-based methods, Edison Academy Magnet School, Edison, NJ
- 2024-26 Alexander Chen, Edge Cases in Crowd Simulation, The Academy of Math, Science and Engineering, Rockaway, NJ
- 2022 Daniel Long, Simulating Agents Moving in Formations, Governor Livingston High School, Berkeley Heights, NJ
- 2022 Sai Voruganti, Estimating Style of 3D Shapes, John P. Stevens High School, Edison, NJ

VIII. Grants and Contracts

- 2022 Learning Deep Representations of Collective Behavior, PI, NJIT-Ben Gurion Seed Grant, 11/2022 to 11/2023, 1 year, \$5,000

- 2022 Understanding Group Motion Dynamics with Reinforcement Learning, PI, NJIT Seed Research, 07/2022 to 06/2023, 1 year, \$7,500
- 2021 Predicting Future and Past Spatial Macroscopic Collective Behaviors, PI, NJIT-Ben Gurion Seed Grant, 03/2021 to 03/2022, 1 year, \$6,000

A. Grant Proposals Pending

- 2026 HCC CoRe: Visualizing Street View for Community Disaster Resilience, NSF, 09/2027 to 09/2030, 3 years, PI, \$400,000 out of 800,000
- 2026 Repulsive Curves for Safe Autonomous Navigation in Dynamic Environments, Amazon, 01/2027 to 12/2027, 1 year, \$50,000
- 2026 CAREER: Simulating Living Systems for Computer Graphics Applications, PI, NSF, 01/2027 to 12/2031, 5 years, \$600,000
- 2026 Physical and Generative AI for Street Scene Synthesis, Sony, 06/2027 to 06/2028, 5 year, \$100,000

IX. Public Recognition

- 2026 Machine Learning and AI Help NJIT Researchers Understand Human Crowd Movement [NJIT News](#)
- 2025 YWCC Student–Faculty Team Wins Best Presentation Award for Ant Swarm Simulation [NJIT News](#), [MSN](#), [TechXplore](#), [Technology Networks](#), [Engtechnica](#), and [other outlets](#)
- 2025 High School STEM Researcher and YWCC Mentor to Present at ACM SIGGRAPH Conference, New Jersey Institute of Technology [News](#)
- 2024 AI Research,
New Jersey Institute of Technology [Media](#)
- 2020 Meet the 2019-2020 New Faculty in Ying Wu College of Computing,
New Jersey Institute of Technology [News](#)
- 2020 Featured New Faculty,
New Jersey Institute of Technology Magazine, [Spr. 2020](#)

X. Service Activities

A. Department

- AY Informatics Department Faculty Candidate Interviewer.
2023-2024
- AY Informatics Department Faculty Candidate Interviewer.
2022-2023
- AY Informatics Department Chair Search Committee.
2022-2023
- AY Informatics Department Faculty Hiring Committee.
2020-2021
- AY Informatics Department Faculty Candidate Interviewer.
2019-2020

B. University

- 2026 NJIT AI Day Breakout Session Speaker
- 2024 Participation in NJIT promotional video on AI Research [Video](#)
- AY NJIT Faculty Senate voting member
2022-2023

C. Government

- 2020,26 Panelist & Reviewer, National Science Foundation
- 2019 [White Paper](#) Contributor, Computing Community Consortium (CCC)
Content Generation for Workforce Training workshop

D. Professional Societies

- 2019- Member: SIAM, ACM, SIGGRAPH, IEEE

E. Conference Organizing and Program Committees

- 2026 Program Committee Member, SIGGRAPH Asia Educator's Forum
- 2026 Program Committee Member, Eurographics Short papers
- 2026 Program Committee Member, ANTS International Conference on Swarm Intelligence
- 2026 Posters co-Chair, ACM SIGGRAPH Symposium on Interactive 3D Graphics and Games (I3D)
- 2026 Technical Program Committee Member, ACM SIGGRAPH Technical Papers
- 2025-26 Program Committee Member, ACM SIGGRAPH Symposium on Interactive 3D Graphics and Games (I3D)
- 2025 Awards Committee Member, ACM SIGGRAPH Motion Interaction and Games
- 2025 Conflict of Interest Coordinator, ACM SIGGRAPH Asia Technical Papers
- 2025 Local Chair, ACM SIGGRAPH Symposium on Interactive 3D Graphics and Games (I3D)
- 2020, 25 Program Committee Member, Computer Animation and Social Agents (CASA)
- 2022-26 Program Committee Member, IEEE International Conference on Computational Photography
- 2022-27 Program Committee Member, IEEE Artificial Intelligence and Virtual Reality
- 2021 Organizing Committee, Workshop on Modeling and Animating Realistic Crowds and Humans (MARCH), IEEE International Conference on Artificial Intelligence & Virtual Reality (AIVR)

F. Peer Reviewing Activity

- 2026 Reviewer, SIGGRAPH Asia Educator's Forum (x3)
- 2026 Reviewer, ACM Transactions on Graphics (x1)
- 2026 Reviewer, IEEE Transactions on Visualization and Computer Graphics (x1)
- 2026 Reviewer, ACM Multimedia, Interactive Art Track (x2)
- 2026 Reviewer, International Conference on Swarm Intelligence (ANTS) (x2)
- 2026 Reviewer, Computer Animation & Virtual Worlds (x1)
- 2026 Reviewer, SIGGRAPH Asia Technical Papers (x1)
- 2026 Reviewer, IEEE Int. Conf. on Computational Photography (x4)
- 2026 Reviewer, Computer Animation and Virtual Worlds (x1)
- 2026 Reviewer, CASA XR (x3)
- 2026 Reviewer, Eurographics Short Papers (x4)
- 2026 Reviewer, SIGGRAPH Technical Papers (x21)
- 2026 Reviewer, ACM I3D (x3)
- 2025 Reviewer, Eurographics State-of-the-Art Reports (x1)
- 2025 Reviewer, IEEE Transactions on Visualization and Computer Graphics (x1)
- 2025 Reviewer, Computer Animation & Virtual Worlds (x1)
- 2025 Reviewer, Springer Nature: Discover Applied Sciences (x1)
- 2025 Reviewer, SIGGRAPH Asia (x2)
- 2025 Paper Shepherd and Area Chair, IEEE Int. Conf. on Computational Photography (x3)

2025 Reviewer, ACM SIGGRAPH Symposium on Interactive 3D Graphics and Games (I3D) (x1)
 2025 Reviewer, SIGGRAPH (x1)
 2024 Reviewer, Computer Animation & Virtual Worlds (x1)
 2024 Reviewer, Eurographics (x1)
 2024 Reviewer, Conference on Human Factors in Computing Systems (CHI) (x1)
 2024 Reviewer, Computer Graphics Forum (x1)
 2024 Reviewer, SIGGRAPH Asia (x1)
 2024 Reviewer, SIGGRAPH (x2)
 2024 Reviewer, Computers & Graphics (x1)
 2023 Reviewer, Eurographics (x1)
 2023 Reviewer, Computer Animation & Virtual Worlds (x1)
 2023 Reviewer, Graphical Models (x1)
 2023 Reviewer, Computer Graphics Forum (multiple)
 2022 Reviewer, IEEE Conference on Virtual Reality (IEEE VR) (x1)
 2022 Reviewer, ACM Symposium on Virtual Reality Software and Technology (VRST) (x1)
 2022 Reviewer, MobileHCI 2022 Late Breaking Results (x1)
 2022 Reviewer, EuroVis poster abstracts (x1)
 2022 Reviewer, IEEE International Symposium on Mixed and Augmented Reality (ISMAR) (x1)
 2021 Reviewer, International Conference on Robotics and Automation (ICRA) (x1)
 2020 Reviewer, Eurographics (x2)
 2020 Reviewer, Computer Animation & Virtual Worlds
 2020 Panelist & Reviewer, National Science Foundation
 2019 Reviewer, Computers & Graphics
 2019 Reviewer, IEEE Access
 2019 PC Member, Computer Animation and Social Agents (CASA)
 2018 Reviewer, Eurographics
 2018 Reviewer, ACM SIGGRAPH Asia
 2018 Reviewer, Elsevier Applied Mathematical Modelling
 2018 Reviewer, Journal of Visualized Experiments

G. Editorial Activity

2022 Guest Editor & Topic Coordinator, Frontiers Virtual Reality

H. Other Activity

2018- Reviewer, UCLA Engineering Undergraduate Scholarships
 2014–2017 UCLA, Graduate Admissions Committee
 2014–2017 UCLA, Computer Science Dept. Graduate Visiting Day Committee
 Led Q&A panels, campus and department overview session
 2014–2016 UCLA, Officer of the Engineering Graduate Student Division Association
 Coordinated student events, volunteering, and outreach. Also in-charge of industry collaboration, and industry sponsored events
 2014–2016 UCLA, Acting President of the Computer Science Graduate Student Committee
 In charge of outreach, industry collaboration and student events. Coordinated company sponsored tea-time events, for recruiting computer science graduate students